Yash Mehta

Software Engineer | AI/ML Developer | Backend & Full-Stack Developer

+91-7408051163 | yash34m6@gmail.com | India

[LinkedIn] | [GitHub]

PROFESSIONAL SUMMARY

Versatile Computer Science and Engineering professional with experience across software development, artificial intelligence, machine learning, backend engineering, and full-stack application development. Proficient in Python, Java, C++, JavaScript, Flask, SQL, and modern machine learning frameworks. Experienced in building end-to-end software applications, developing REST APIs, integrating databases, and implementing intelligent systems. Developed 10+ technical projects, contributed to research presented at IIT Roorkee, and co-authored publications in Springer Nature and IEEE proceedings. Seeking opportunities to contribute to software engineering and technology-driven product development.

TECHNICAL SKILLS

- **Programming**: Python, C++, Java, JavaScript
- **Software Development:** Object-Oriented Programming, Application Development, REST APIs, Backend Engineering
- **Frontend:** HTML, CSS, JavaScript, Chart.js
- **Backend Frameworks:** Flask, Java Servlets, JSP
- **Databases:** MySQL, MongoDB
- **Artificial Intelligence:** TensorFlow, Scikit-learn, CNN, LSTM, GRU, Vision Transformers
- **Data Science:** Pandas, NumPy, ARIMA, Prophet, Feature Engineering, Model Evaluation
- **Computer Vision & NLP:** OpenCV, TF-IDF, Sentiment Analysis
- **Tools & Platforms:** Git, GitHub, Jupyter Notebook, Google Colab, Gemini API

SELECTED DATA & ANALYTICS PROJECTS

FIR Legal Assistant - LexHack | AI-Powered Software Application | 2025

- Developed an FIR lodging assistance application using Flask, Gemini AI, and Google Maps API.
- Integrated intelligent assistance and location-aware application functionality.
- Received the Innovation & Impact award from the hackathon evaluation panel.

Fake News Detection Using Machine Learning | 2025

- Developed a Flask-based machine learning application using TF-IDF, Random Forest, and sentiment analysis.
- Achieved 87% validation accuracy.
- Presented research at SocProS-25, IIT Roorkee.

- Research subsequently published through Springer Nature.

Amazon Stock Price Prediction | Predictive Analytics | 2025

- Implemented financial forecasting using LSTM, GRU, ARIMA, and Transformer architectures.
- Achieved RMSE of 0.03285 in forecasting experiments.
- Designed a custom multivariate TimeSeriesTransformer.

Fog Detection Using Hybrid Deep Learning Models | 2025

- Developed a hybrid computer vision approach involving CNN and Vision Transformer architectures.
- Worked with the FRIDA2 dataset and integrated OpenCV.
- Reported a 15% improvement in accuracy on real-world haze datasets.
- Published research in IEEE ICCCA proceedings.

Retail Analytics Dashboard - Vishleshan | Full-Stack Development | 2023

- Developed a retail dashboard using Flask, MySQL, and Chart.js.
- Implemented sales analytics and data visualization.
- Presented the project to 200+ industry experts at Bennett University.

Room Allocation Tool | Java & MySQL | 2023

- Developed a desktop application for automated student room allocation.
- Implemented MySQL database integration and allocation workflows.
- Supported 100+ students and reduced manual effort by approximately 95%.

Depression Detection via Facial Cues | Computer Vision | 2024

- Developed a CNN-based facial image classification prototype using OpenCV.
- Achieved 82% classification accuracy in project evaluation.
- Integrated the model into a Flask application.

PROFESSIONAL EXPERIENCE

Software Intern | MNNIT Allahabad | 2024

- Developed REST APIs for academic services using Python and Flask.
- Improved backend throughput by 20%.
- Collaborated with team members during Agile development sprints.

RESEARCH EXPERIENCE

Fake News Detection | IIT Roorkee | 2025

- Developed an ML classification pipeline integrating TF-IDF, Random Forest, and sentiment analysis.
- Implemented a Flask-based research demonstration.
- Presented at SocProS-25.

Blockchain in Autonomous Vehicles | Bennett University | 2024

- Simulated blockchain protocols for autonomous vehicle communication.
- Evaluated spoofing-related security risks in experimental environments.
- Reported a 35% reduction in spoofing risks under tested conditions.

RESEARCH PUBLICATIONS

1) Fake News Detection Using Machine Learning and Sentiment Analysis Integrated in a Flask Web Application.

In: Pant, M., Deep, K., Nagar, A.K. (eds.), Real-World Applications of Machine Learning. Springer, Singapore, 2026.

DOI: https://doi.org/10.1007/978-981-95-7134-5_23

2) Fog Detection Using Hybrid Deep Learning Models: A Multi-Modal CNN-LSTM Approach.

IEEE ICCCA, 2025.

DOI:10.1109/ICCCA66364.2025.11325690

CERTIFICATIONS

- NVIDIA - Fundamentals of Deep Learning
- IBM - IBM Machine Learning Professional Certificate
- Google Cloud - Introduction to Generative AI
- IBM - Generative AI Fundamentals
- IBM - AI Foundations for Business
- Google - Google Data Analytics Professional Certificate
- IIT Delhi - Quantum Computing & QKD

EDUCATION

B.Tech. - Computer Science & Engineering

Bennett University | 2022–2026 | CGPA: 8.26

Senior Secondary (Class XII) - CBSE | 2021 | 92.4%

Secondary (Class X) - CBSE | 2019 | 85.8%

Maharshi Patanjali Vidya Mandir